

AMIT MANDAL



📍 West Bengal ✉ marchamit07@gmail.com ☎ 8637814826 in LinkedIn 📄 GitHub 🌐 Portfolio

M.Tech Computer Science graduate from NIT Durgapur with a strong foundation in Software Development, Full Stack & AI/ML. Skilled in C++, Python, MERN, RAG & AI Agents seeking Tech. roles(SDE, Full-Stack, AI, Automation) to contribute while building impactful Software Solutions.

Education

National Institute of Technology(NIT) Durgapur <i>M.Tech. in Computer Science and Engineering</i> · CGPA: 8.49	<i>Aug 2024 - May 2026</i>
Cooch Behar Government Engineering College <i>B.Tech. in Computer Science and Engineering</i> · CGPA: 8.89	<i>Nov 2020 - Jul 2024</i>
Akrurmoni Coronation Institution <i>HS (12th) in Science</i> · Percentage : 90%	<i>May 2017 - Jun 2019</i>

Projects

JobTrackerAI – AI Powered Smart Job Application Management Platform [🔗](#)

- Built a production deployed MERN platform with JWT authentication, drag-and-drop Kanban board and real-time job pipeline tracking, reducing manual application management effort significantly.
- Integrated Google Gemini API with PDF parsing to deliver AI driven ATS match scoring, resume gap analysis and personalized cover letter generation, enabling smarter and faster job applications.
- Engineered a scalable REST API backend (Node.js, Express.js, MongoDB Atlas) with async Mongoose hooks and cloud deployment on Render & Vercel, ensuring high availability and seamless frontend-backend integration.

Deep Packet Inspection (DPI) Engine [🔗](#)

- Engineered a high-performance Deep Packet Inspection (DPI) engine in C++ with multi-threaded packet processing for scalable traffic analysis.
- Built a protocol-aware parsing pipeline (Ethernet → IP → TCP/UDP) with deep packet inspection and SNI extraction for traffic classification across web platforms such as YouTube, Facebook & similar services.
- Designed a rule-based filtering system for domain & IP-level access control to enforce scalable network policies.

Water Quality Index Prediction [🔗](#)

- Developed a Machine Learning model for Water Quality Index (WQI) prediction using CPCB data, applying feature engineering and data analysis with Random Forest, SVM & other ML Models, achieving 93%+ prediction accuracy.

GradeSync – CGPA&Percentage Converter [🔗](#)

- Developed GradeSync, a web based CGPA ↔ Percentage conversion platform supporting multiple university-specific grading formulas for accurate and instant academic score conversion.

Training & Certifications

Deloitte <i>Data Analytics Job Simulation / Virtual Internship.</i> 🔗	<i>May 2025 - July 2025</i>
◦ Analyzed structured datasets to identify anomalies, trends and fraud indicators, generating insights to support digital forensic investigations, risk assessment and data-driven decision-making.	
Accenture <i>Developer Virtual Experience.</i> 🔗	<i>June 2023 - Aug 2023</i>
◦ Improved system architecture for cloud migration by implementing IAM policies, debugging, testing and UAT while following secure SDLC practices to enhance security and system reliability.	

Skills

Languages: C++, C, Python, JavaScript, SQL, HTML, CSS
Databases: MySQL, MongoDB, MongoDB Atlas
Frameworks & Libraries: React.js, Tailwind, FastAPI, Pandas, NumPy, Streamlit, JWT.
Tools & Platforms: Express.js, Node.js, Docker, Postman, Git, GitHub, Linux, VSCode, Jupyter, Kaggle.
AI & Emerging Tech: RAG, LangChain, LangGraph, LlamaIndex, Hugging Face, Ollama, AI Agents.
Coursework: DSA · OS · OOPs · CN · DBMS
Soft Skills: Analytical Thinking · Problem Solving · Communication · Adaptability · Team Collaboration

Achievements & Interests

- Solved 150+ **LeetCode** problems, strengthening data structures, algorithms, and problem-solving skills. [🔗](#)
- Qualified **GATE** in 2024 & **WebJEE** in 2020.
- Selected to represent Malda district as a **Wicketkeeper-Batsman** at the inter-district level Cricket Tournament.
- Passionate about **Travel** and **Cinematography**, exploring new cultures, connecting with people and blending creative vision with technical storytelling.